

4	(a) for a system (of interacting bodies) the <u>total</u> momentum remains constant provided there is no <u>resultant</u> force acting (on the system)	M1 A1	[2]
	(b) (i) total momentum = $m_1v_1 + m_2v_2$ = $0.4 \times 0.65 + 0.6 \times 0.45$ = $0.26 + 0.27 = 0.53 \text{ Ns}$	C1 C1 A1	[3]
	(ii) $0.53 = 0.4 \times 0.41 + 0.6 \times v$ $v = 0.366 / 0.6 = 0.61 \text{ m s}^{-1}$	C1 A1	[2]
	(iii) $\text{KE} = \frac{1}{2}mv^2$ total initial KE = $\frac{1}{2} \times 0.4 \times (0.65)^2 + \frac{1}{2} \times 0.6 \times (0.45)^2$ = $0.0845 + 0.06075 = 0.15 (0.145) \text{ J}$	C1 C1 A1	[3]
	(c) check relative speed of approach equals relative speed of separation or: total final kinetic energy equals the total initial kinetic energy	B1	[1]
	(d) the forces on the two bodies (or on X and Y) are equal and opposite time same for both forces <u>and</u> force is change in momentum/time	B1 B1	[2]